

Homework 1 | STAT 330

Simple Linear Regression

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Data and Description

Energy can be produced from wind using windmills. Choosing a site for a wind farm (i.e. the location of the windmills), however, can be a multi-million dollar gamble. If wind is inadequate at the site, then the energy produced over the lifetime of the wind farm can be much less than the cost of building the operation. Hence, accurate prediction of wind speed at a candidate site can be an important component in the decision to build or not to build. Because the energy produced is proportional to the square of the wind speed, even small errors in wind speed prediction can lead to significant impacts.

A potential approach for predicting wind speed at a candidate site is to use wind speed data from a nearby reference site. A reference site is a nearby location where the wind speed is already being monitored and should, theoretically, be similar to the candidate site. If the reference sites turn out to be good predictors of the candidate sites, leveraging the information from the reference site would allow windmill companies to estimate the wind speed at the candidate sites without going through a costly and lengthy data collection period.

The Windmill data set contains measurements of wind speed (in meters per second m/s) at **candidate sites (CSpd)** (**column 1**) and at corresponding **reference sites (RSpd)** (**column 2**) for 1,116 sites. Download the Windmill.txt file from Canvas (which can be found in Files -> Data Sets), and put it in the same folder as this quarto file.

0. Replace the text “< PUT YOUR NAME HERE >” (above next to “author:”) with your full name.

1. Briefly explain why simple linear regression could be a useful tool for this problem.

Simple linear regression would be a useful tool for this problem because we want to investigate the nature of the relationship between the wind speeds at candidate sites and their corresponding reference sites. This means that we have one variable and one response, making simple linear regression appropriate.

2. Read in the data set into an object named “wind”. Display the first 6 rows of the data.

```
windmill_data <- read.table("Windmill.txt", header = TRUE)
head(windmill_data)
```

	CSpd	RSpd
1	6.9	5.9666
2	7.1	7.2176
3	7.8	7.9405
4	6.9	6.0174
5	5.5	6.1646
6	3.1	1.7687

3. What is the response/outcome variable in this situation? (Think about which variable makes the most sense to be the response.)

The wind speed at the candidate sites, or the CSpd column.

4. What is the explanatory variable in this situation?

The wind speed at the reference sites, or the RSpd column.

5. Create a scatterplot of the data with variables on the appropriate axes. Add descriptive axis labels with appropriate units. Display the plot.

```
#Got some help here for creating a scatter plot using ggplot from chatgpt

library(ggplot2)

# Create scatter plot with labels
plot <- ggplot(windmill_data, aes(x = RSpd, y = CSpd)) +
  geom_point(color = "blue", size = 1) +
  labs(title = "The Relationship Between Wind Speeds at Reference Sites and Candidate Sites",
       x = "Reference Site Wind Speed (m/s)",
       y = "Candidate Site Wind Speed (m/s)") +
  theme_minimal()

# Print the plot
print(plot)
```